UX-0009 — Butlerly Accessibility Standard
Version 1.0

Purpose
This document defines the mandatory accessibility standard for Butlerly Version 1 across smartphone, tablet, and desktop. It governs inclusive interaction, visual presentation, content, assistive-technology support, keyboard and switch access, large text, contrast, motion, audio, cognitive load, AI explanations, cross-device continuity, testing, governance, and release acceptance.

1. Accessibility Objective
Butlerly must enable people with diverse visual, motor, hearing, speech, cognitive, neurological, and situational needs to complete the same essential product tasks with comparable privacy, control, dignity, and confidence.

Accessibility is not an optional enhancement, alternate mode, or post-release correction. It is part of the definition of every screen, component, flow, state, and acceptance criterion.

The product must:
- Support independent use wherever technically possible.
- Preserve equal access to capture, review, search, correction, export, device management, privacy, and recovery.
- Avoid requiring a single sensory, motor, or input method.
- Remain usable with platform accessibility settings enabled.
- Make AI-generated content, confidence, evidence, and uncertainty understandable.
- Keep security and privacy protections accessible without weakening them.
- Provide durable recovery when transient feedback is missed.

2. Conformance Target
Version 1 targets WCAG 2.2 Level AA for applicable user interfaces and content, together with the current accessibility guidance of each supported native platform.

When requirements differ:
- The stricter user-protective requirement applies when practical.
- Native platform behavior is preferred when it provides stronger assistive-technology support.
- Any exception requires documented rationale, user impact, mitigation, owner, and retirement plan.

Legal compliance is a minimum. Butlerly's product standard is successful task completion, not checklist completion alone.

3. Governing Principles

3.1 Equivalent Outcomes

Users must be able to reach the same meaningful outcome through accessible paths. An alternative interaction may differ in presentation but must not remove capability, privacy, or control.

3.2 Multiple Input Methods
Core tasks must not depend exclusively on touch, pointer, drag, hover, camera, voice, color, sound, or precise gestures.

3.3 Clear Structure
Information hierarchy, headings, groups, labels, and reading order must remain understandable visually and through assistive technology.

3.4 Visible and Programmatic State
Selection, focus, validation, confidence, synchronization, offline status, loading, and completion must be exposed visually and programmatically.

3.5 Reversible and Recoverable
Undo, version history, Recently Deleted, and persistent review queues protect users who miss transient feedback or make accidental actions.

3.6 Privacy Without Barriers
Authentication, consent, device trust, export, and deletion flows must remain accessible. Accessibility must not require disclosure of more private information than other use paths.

3.7 Calm, Direct Communication
Accessibility copy must be specific, non-blaming, concise, and action-oriented. Errors explain what failed, what was preserved, and what happens next.

4. Supported Access Technologies
Butlerly Version 1 must be designed and tested with applicable platform technologies, including:
- Screen readers.
- Voice control.
- Switch control and switch scanning.
- Full keyboard access.
- Pointer and alternative pointing devices.
- Platform zoom and magnification.
- Dynamic Type or equivalent text scaling.
- High contrast and increased-contrast modes.
- Reduced motion and reduced transparency.
- Color filters and color-vision accommodations.
- Bold text and larger pointer settings.
- Dictation and voice input where supported.
- Captions and transcripts for any media containing speech.

5. Semantic Structure
Every interactive element must expose an accurate accessible name, role, value, state, and relationship.

Requirements:
• Headings form a logical hierarchy.
• Lists, groups, tables, forms, navigation, dialogs, alerts, and landmarks use appropriate semantics.
• Visual grouping is represented programmatically.
• Labels remain associated with controls.
• Required, invalid, expanded, selected, checked, pressed, busy, and disabled states are exposed.
• Custom controls must match native semantic behavior before release.
• Decorative images and icons are hidden from assistive technology.
• Meaningful images have concise descriptions or equivalent text.
• Repeated controls have context-specific names, such as “Edit category” rather than “Edit.”

6. Reading and Focus Order
Reading order and focus order must follow the task’s logical sequence, not implementation order or visual coordinates alone.

Rules:
• Focus starts at the new page title or the first meaningful task element.
• Dialogs and sheets trap focus while open and return focus to the initiating control when closed.
• Newly inserted validation or review content is announced without unexpectedly moving focus.
• Focus is not reset during background synchronization, AI streaming, or list refresh.
• Multi-pane layouts expose a predictable order and clear pane names.
• Hidden or offscreen content is not focusable.
• Destructive confirmation places initial focus on the least destructive reasonable control.
• Deep links restore focus near the intended task.

7. Keyboard Access
Every core task must be completable without touch or pointer on tablet and desktop, and wherever platform support allows on smartphone.

Requirements:
• All controls are reachable in logical order.
• Focus indicators are always visible.
• No keyboard trap exists.
• Standard platform keys retain expected behavior.
• Menus, dialogs, tables, lists, tabs, disclosure controls, and segmented controls follow platform conventions.
• Escape or platform-equivalent dismisses temporary surfaces when safe.
• Enter and Space activate controls according to native expectations.

• Arrow keys navigate composite controls appropriately.
• Keyboard shortcuts are discoverable and never the only path.
• Destructive shortcuts require confirmation or an immediate durable recovery path.

8. Touch and Pointer Targets
Interactive targets must meet platform minimums and remain usable for people with limited dexterity.

Rules:
• Primary touch targets should be at least 44 by 44 points on Apple platforms or the current platform-equivalent minimum.
• Visual size may be smaller only when the hit area meets the minimum and does not overlap nearby targets.
• Adjacent destructive and safe actions require sufficient separation.
• Precision dragging is never the only way to reorder, link, merge, or transfer.
• Hover content must also be available through focus, tap, or explicit disclosure.
• Pointer targets preserve visible hover and focus distinctions.
• Long press and multi-finger gestures always have visible alternatives.

9. Typography and Text Scaling
Butlerly uses native system fonts and supports platform text scaling.

Requirements:
• Body content scales to the platform's largest supported accessibility sizes where practical.
• Text reflows rather than clipping, overlapping, or hiding essential controls.
• Horizontal scrolling is not required for normal reading content.
• Fixed-height text containers are avoided.
• Buttons adapt to multiline labels when needed.
• Truncation never removes critical amounts, dates, device names, privacy consequences, or action meaning.
• Tables provide alternate stacked or detail views when large text makes the table unusable.
• Line length remains readable on wide desktop layouts.
• Text embedded in images is prohibited unless a full accessible equivalent is provided.

10. Color and Contrast
Color is never the only way to communicate meaning.

Requirements:
• Text and meaningful icons meet applicable AA contrast.
• Large text follows the applicable large-text threshold.
• Focus indicators, control boundaries, selected states, and meaningful graphics meet non-text contrast requirements.
• Error, warning, success, security, AI, and synchronization states include text or shape cues.
• Financial gains and losses never rely solely on green and red.

- Confidence always includes High, Medium, or Low text.
- Light, dark, system, and high-contrast modes are tested independently.
- Disabled controls remain identifiable without becoming unreadable.
- Placeholder text is not used as the sole label or instruction.

11. Motion, Animation, and Flashing
Motion communicates relationship and progress but must not create discomfort or block comprehension.

Rules:
- Reduced Motion removes nonessential animation, parallax, zooming, decorative generation effects, and spatial transitions.
- Essential state changes use immediate updates or subtle fades when motion is reduced.
- No content flashes at unsafe frequencies.
- Infinite decorative animation is prohibited.
- Processing indicators do not imply completion.
- Auto-advancing content is avoided; when present, users can pause, stop, or control it.
- Haptics and sound are supplemental, never required.

12. Audio, Speech, and Voice
Any audible information must have a visual and programmatic equivalent.

Requirements:
- Product sounds are optional and do not carry unique meaning.
- Spoken media includes captions and transcripts.
- Voice input is optional and has text, touch, or keyboard alternatives.
- Dictated text is reviewable before consequential use.
- Voice-control names match visible labels where possible.
- Sensitive content is not spoken unexpectedly.
- Screen-reader announcements avoid repeating private data unnecessarily.

13. Forms and Validation
Forms must minimize memory, preserve work, and make correction straightforward.

Requirements:
- Every field has a persistent visible label.
- Instructions appear before they are needed.
- Required fields are clearly identified in text.
- Input purpose and format are programmatically available where supported.
- Validation occurs at an appropriate time and does not interrupt typing.
- Error messages identify the field, problem, and correction.
- An error summary links to affected fields in long forms.
- User input is preserved after errors, network loss, or provider failure.
- AI-proposed values use the same accessible editing controls as user-entered values.

• Source evidence can be opened without losing the form state.

14. Navigation and Orientation
Navigation must remain stable, predictable, and perceivable.

Requirements:
• Major destinations use consistent names and positions.
• Current destination and selection are exposed programmatically.
• Screen titles describe the current context.
• Back behavior preserves filters, selection, and position where practical.
• Orientation changes preserve task state.
• Tablet compact and expanded layouts preserve the selected item.
• Desktop multi-window use includes clear window titles.
• AI does not replace accessible navigation.

15. Lists, Tables, and Data
Financial and record data must remain understandable through visual and nonvisual navigation.

Lists:
• Each row has one clear primary target.
• Trailing actions have specific names.
• Status and metadata are exposed in meaningful order.
• Reordering and multi-selection have non-drag alternatives.

Tables:
• Row and column headers are programmatically associated.
• Sort state is announced.
• Filters and active filter counts are exposed.
• Keyboard users can navigate without entering every interactive cell unnecessarily.
• Amounts, dates, and statuses have understandable text equivalents.
• Complex tables provide row-detail or alternate list views.
• Batch actions apply only to explicit selection and announce the selected count.

16. Images, Receipts, and Source Evidence
Source artifacts are central to Butlerly and require accessible alternatives.

Requirements:
• Receipt and document images expose useful descriptions.
• Extracted text is available as an accessible alternative when present.
• Users can navigate extracted fields and their source relationships.
• Zoom, pan, and rotation have accessible controls.
• Visual bounding boxes are not the only indication of field-source mapping.
• Low-quality images are described honestly.
• Original source and interpreted record are distinguishable.

• Sensitive image previews are hidden from task switchers or notifications where platform controls allow.

17. AI Accessibility
AI interactions must remain understandable, controllable, and operable through assistive technologies.

Every material AI result must expose:
• What Butlerly believes.
• Why it believes it, using concise evidence.
• Confidence or uncertainty.
• What Butlerly did or recommends.
• What the user can do next.

Rules:
• Observation, inference, and recommendation are distinguishable in text.
• Confidence is never color-only.
• Streaming responses do not trigger repeated disruptive announcements.
• A completed answer is announced once with a concise status.
• Suggested prompts are real controls with accessible names.
• AI proposals are editable and correctable.
• Evidence links are keyboard and screen-reader accessible.
• External-processing consent is fully accessible.
• AI unavailability never blocks access to locally stored records.
• The Assistant does not claim an action succeeded until the system confirms it.

18. Notifications, Alerts, and Live Regions
Notifications must be timely without becoming disruptive.

Requirements:
• Security, destructive, and blocking errors use persistent accessible alerts.
• Routine saves and synchronization do not create repeated announcements.
• Background updates use polite announcements only when the user benefits.
• Urgent alerts do not interrupt text entry unnecessarily.
• Toasts never contain the only copy of information needed later.
• Every transient Undo has a durable history or recovery path where consequences matter.
• Lock-screen notifications avoid merchant names, amounts, and document contents by default.

19. Authentication, Privacy, and Security
Security flows must be accessible without reducing protection.

Requirements:
• Authentication prompts explain the protected action.
• Biometric authentication has the platform’s accessible fallback.

• Consent choices are individually named and no approval is preselected.
• Device fingerprints have human-readable verification alternatives.
• QR pairing has a manual or verification-phrase alternative.
• Device revocation identifies the device by understandable name and platform.
• Export and deletion scopes are reviewable before confirmation.
• Recovery codes or secrets are not exposed through screen-reader output unless the user explicitly requests access and the platform can protect it.
• Timeout warnings provide enough time and extension controls when security policy permits.

20. Cross-Device Accessibility
A user may use different accessibility methods on different devices.

Rules:
• Platform accessibility preferences remain device-specific by default.
• Data and task continuity do not assume identical input methods.
• Continue Recent Work restores semantic task context, not a visual-only position.
• Pairing, synchronization, conflict resolution, and recovery are accessible on all supported device classes.
• Device names and states are understandable without icons alone.
• Same-field resolutions and merges are available in accessible version history.
• Revocation and lost-device flows remain keyboard and screen-reader operable.

21. Cognitive Accessibility
Butlerly must reduce unnecessary memory, ambiguity, and decision burden.

Requirements:
• One clear primary action per decision area.
• Complex tasks are divided into meaningful steps.
• Progress and remaining work are visible.
• Terminology is consistent.
• Instructions use plain language.
• Destructive consequences are concrete.
• Time limits are avoided unless security requires them.
• Users can defer nonblocking review items.
• Drafts and input are preserved.
• Empty states explain what the area is and what to do next.
• AI asks one focused question at a time.
• The interface avoids shame, pressure, gamification, and engagement manipulation.

22. Error and Recovery Standard
Every recoverable error must state:
• What failed.
• What was preserved.
• Whether private data left the device when relevant.

• What remains usable.
• What the user can do next.
• Whether retrying is safe.

Errors must be announced, visually identifiable, and reachable through keyboard and screen reader. Error codes may appear in diagnostics but never replace plain-language explanation.

23. Localization and Accessibility

Localization must preserve accessibility semantics.

Requirements:
• Text expansion is supported.
• Accessible names are localized, not assembled from fragments.
• Dates, times, amounts, currencies, and numbers follow locale.
• Right-to-left layout preserves logical reading and focus order.
• Directional icons mirror appropriately.
• Translated terminology remains consistent with the Butlerly glossary.
• Screen-reader pronunciation is reviewed for key financial and product terms.
• Mixed-language source documents remain navigable.

24. Platform-Specific Baselines

24.1 iPhone and iPad

Test with VoiceOver, Voice Control, Switch Control, Dynamic Type, Zoom, Bold Text, Increase Contrast, Reduce Motion, Reduce Transparency, color filters, external keyboard, and pointer where applicable.

Use native accessibility APIs, traits, actions, rotors, focus management, and content-size categories. Custom gestures require visible alternatives.

24.2 Desktop

Test with the platform screen reader, full keyboard access, voice control, zoom, high contrast, reduced motion, larger text, pointer alternatives, and window-management accessibility.

Desktop must provide visible focus, logical shortcut behavior, accessible tables, resizable panes, and usable content at increased scaling.

25. Component Definition Requirements

Every production component must document:
• Accessible name strategy.
• Role and state.
• Reading order.
• Focus behavior.
• Keyboard behavior.

- Touch target.
- Text scaling behavior.
- Contrast requirements.
- Reduced-motion behavior.
- Screen-reader announcements.
- Error and disabled states.
- Platform variations.
- Automated and manual test coverage.

A component is not complete until these behaviors are implemented and tested.

26. Design Review Checklist

Before engineering handoff, design must verify:

- Complete keyboard and non-pointer path.
- Logical reading and focus order.
- Large-text reflow.
- Light, dark, and high-contrast states.
- Non-color status communication.
- Touch-target spacing.
- Error identification and recovery.
- Screen-reader labels and grouping.
- Reduced-motion behavior.
- Accessible alternatives for drag, hover, QR, camera, voice, and gestures.
- AI confidence, evidence, consent, and correction.
- Privacy-safe notifications and previews.

27. Engineering Acceptance

Engineering must:

- Prefer native controls and accessibility APIs.
- Add semantics during component implementation, not afterward.
- Include accessibility states in unit and UI tests.
- Prevent regressions through automated checks where reliable.
- Test real assistive-technology behavior manually.
- Preserve focus during asynchronous updates.
- Expose loading, completion, selection, and validation state.
- Avoid inaccessible custom controls without approved justification.
- Document known platform limitations.

28. Testing Strategy

Accessibility testing includes four layers:

Layer 1 — Automated

Contrast, missing labels, invalid roles, focusability, target size where detectable, and structural issues.

Layer 2 — Manual Expert Review
Keyboard, screen reader, large text, zoom, reduced motion, high contrast, forms, dialogs, tables, and dynamic updates.

Layer 3 — Task-Based Assistive-Technology Testing
Complete real journeys such as capture, review, search, correction, device pairing, export, revocation, and recovery.

Layer 4 — Usability Testing with Disabled Participants
Representative users with varied disabilities and assistive technologies validate real comprehension, effort, privacy, and confidence.

Automated tests never replace manual and user testing.

29. Required Version 1 Test Scenarios
• First launch and privacy explanation with screen reader.
• Receipt capture without relying on camera alignment alone.
• Manual entry with large text.
• Review and correct an AI proposal using keyboard only.
• Understand confidence without color.
• Inspect receipt source with screen reader.
• Search and filter records with voice control.
• Navigate a desktop table with keyboard and screen reader.
• Pair a device using QR alternative.
• Complete initial synchronization with status announcements.
• Resolve a same-field conflict.
• Revoke a device with authentication fallback.
• Work offline and understand pending changes.
• Recover from provider failure.
• Export selected data.
• Delete and restore a record.
• Use dark mode, high contrast, and reduced motion.
• Use the largest supported text size.
• Complete core tasks in a right-to-left test locale.

30. Severity and Release Policy
Accessibility defects use these severities:

Blocker
A core task cannot be completed by a supported assistive-technology user; security or privacy is inaccessible; data loss risk exists; or no equivalent path exists.

Critical

A major task is technically possible but unreliable, misleading, or excessively difficult.

Major
A secondary task, state, or component has a meaningful accessibility barrier.

Minor
A localized issue causes limited friction without preventing task completion.

Release rules:
• No known Blocker defects.
• No unmitigated Critical defects in core Version 1 journeys.
• Major defects require an owner and committed correction plan.
• Exceptions require Product, Design, Engineering, and accessibility review.

31. Accessibility Metrics
Track privacy-preserving measures such as:
• Core task completion with screen reader.
• Core task completion with keyboard only.
• Large-text layout pass rate.
• Automated accessibility test pass rate.
• Accessibility defects by severity and age.
• Time to resolve Blocker and Critical issues.
• Percentage of components with complete accessibility contracts.
• Assistive-technology usability satisfaction.
• Error recovery success.
• Accessibility-related support themes.

Do not collect disability status or assistive-technology use without explicit purpose, consent, and privacy review.

32. Governance
Design owns inclusive interaction requirements and component behavior. Engineering owns correct implementation and platform integration. Product owns prioritization without waiving the standard. QA owns independent verification. Accessibility specialists and disabled users must be included at meaningful review points.

Any exception must document:
• Requirement not met.
• Affected users and tasks.
• Reason.
• Temporary alternative.
• Risk.
• Owner.
• Target removal date.

33. Dependencies
UX-0009 depends on:
• UX-0001 — Butlerly Experience Blueprint.
• UX-0002 — Butlerly Version 1 User Journey Map.
• UX-0003 — Butlerly Information Architecture and Navigation Model.
• UX-0004 — Butlerly Core User Flows.
• UX-0005 — Butlerly Screen and State Inventory.
• UX-0006 — Butlerly AI Interaction and Explainability Specification.
• UX-0007 — Butlerly Design System Foundations.
• UX-0008 — Butlerly Cross-Device Experience and Continuity.
• Current supported-platform accessibility guidance.

UX-0009 governs:
• UX-0010 — Low-Fidelity Wireframe Specification.
• UX-0011 — Design Patterns and Component Library.
• UX-0012 — Usability Testing and UX Metrics.
• Engineering accessibility acceptance criteria.
• Design QA and release readiness.

34. Acceptance Criteria
This standard is complete when:
• Core journeys have equivalent accessible outcomes.
• Screen-reader, keyboard, switch, voice, large-text, contrast, and reduced-motion requirements are explicit.
• AI, privacy, security, synchronization, and recovery interactions are covered.
• Smartphone, tablet, and desktop baselines are defined.
• Every component has a testable accessibility contract.
• Required manual and task-based test scenarios are documented.
• Defect severity and release rules are enforceable.
• Engineering and QA can derive implementation and regression tests.
• Accessibility is integrated into design, implementation, and release governance.

BDS-0000 Conformance
UX-0009 remains the authoritative accessibility standard where its requirements are stricter, while BDS-0000 — Butler Design System v1.0 supplies the concrete visual tokens, typography values, spacing/radius, responsive breakpoints, component contracts, theming, and Flutter implementation constraints. Finance V1 visible primary navigation is Home, Transactions, Review, Search, and Settings; Assistant is optional/non-primary. Cross-device/device-trust accessibility requirements remain valid future-design requirements but do not make synchronization an MVP dependency. UI-authored accessible names and content follow the selected application language; user-entered/imported/scanned/source data preserves its original language/script by default, including mixed-language content. Original financial

amount/currency remains authoritative and any reference conversion is exposed as a separate value.